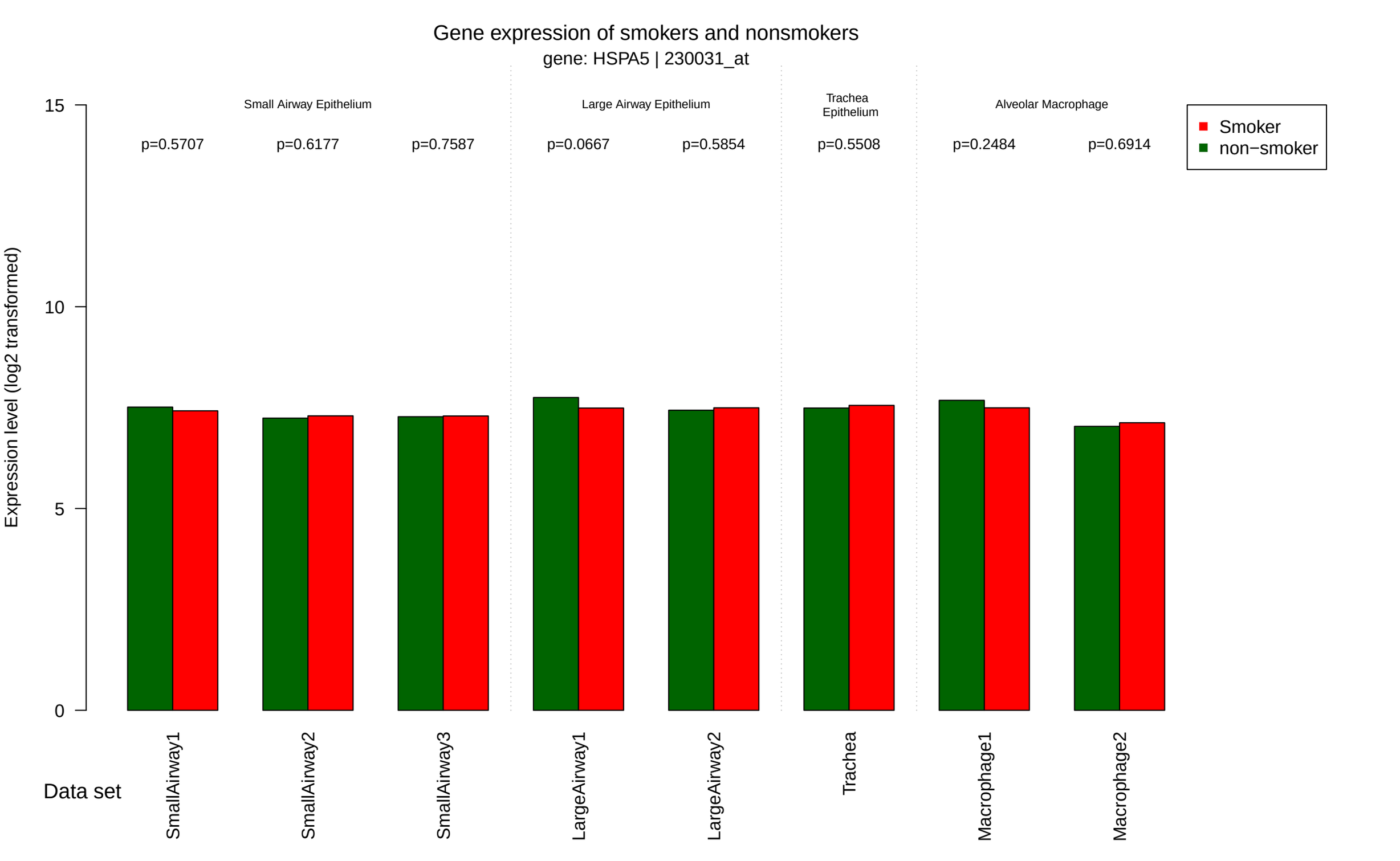
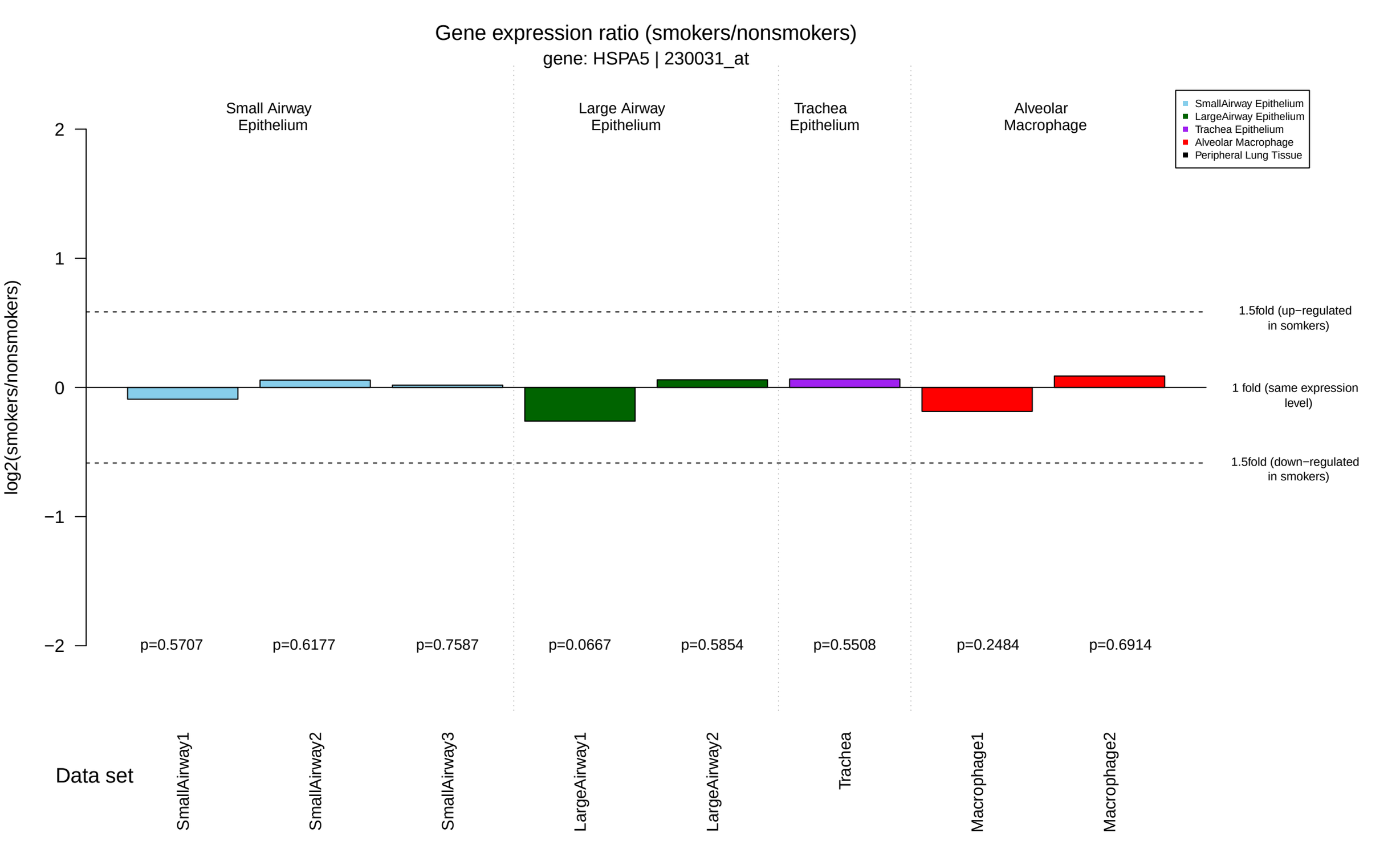
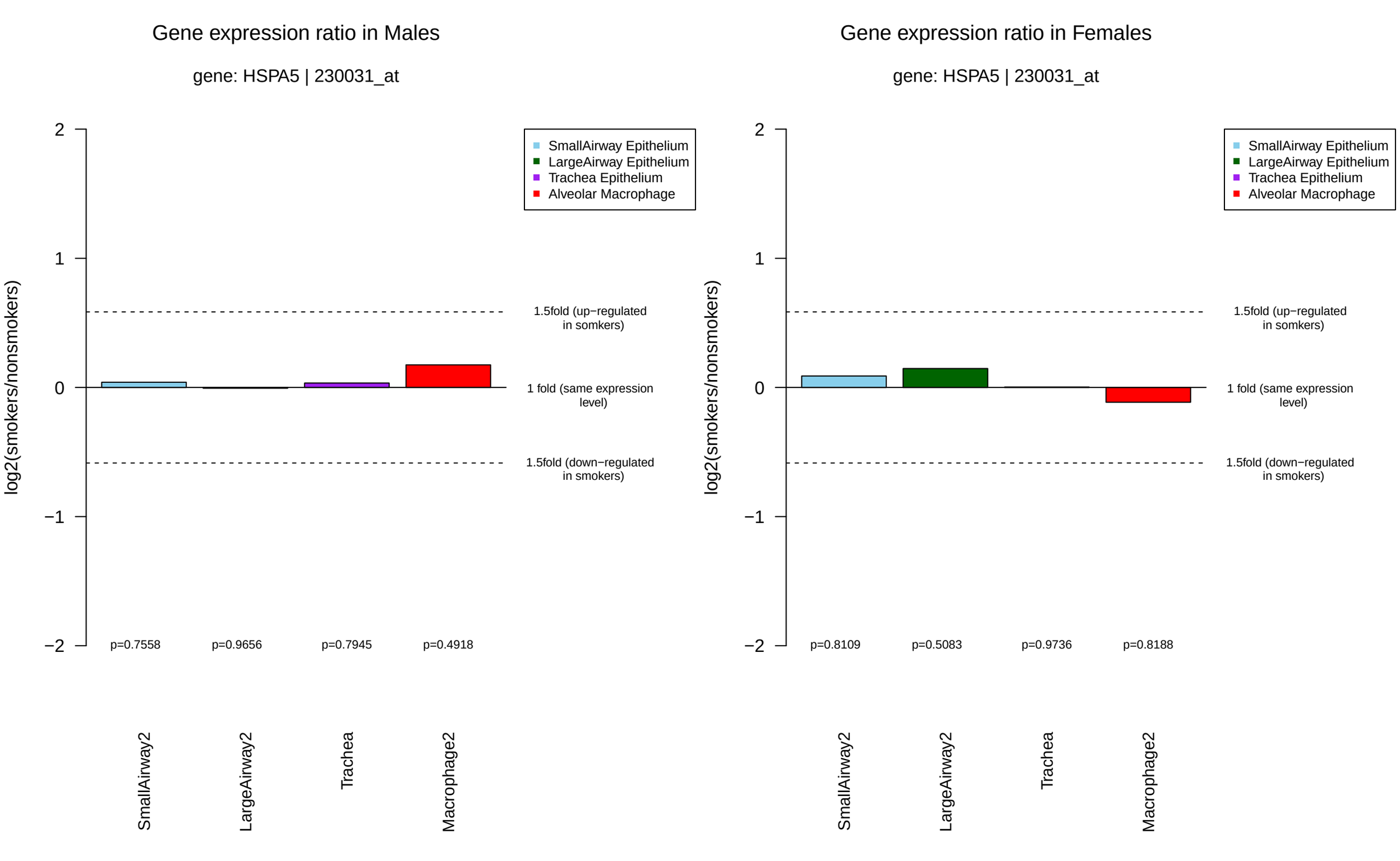
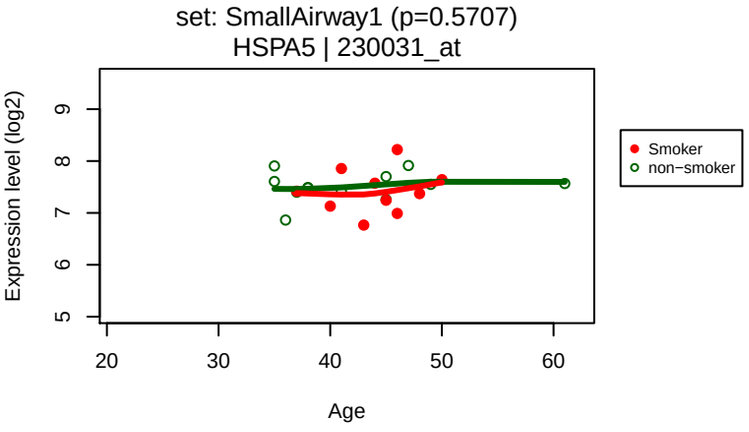
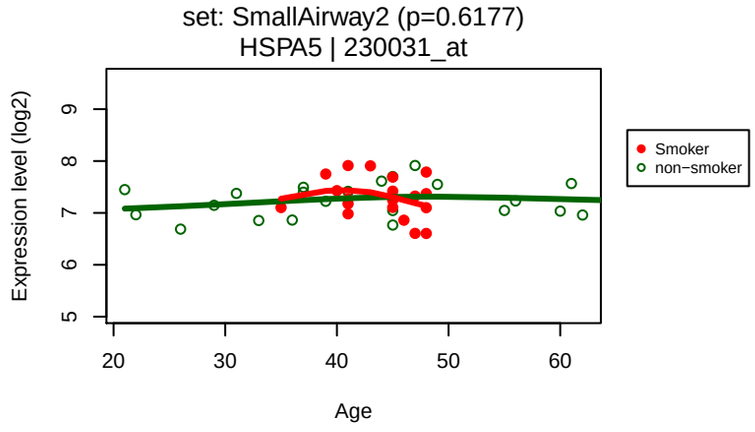
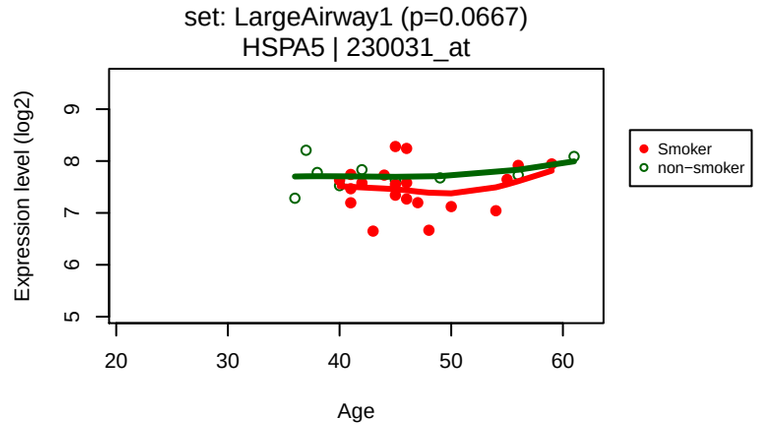

Gene expression of smokers and nonsmokers
gene: HSPA5 | 230031_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5707
p=0.6177
p=0.7587
p=0.0667
p=0.5854
p=0.5508
p=0.2484
p=0.6914
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

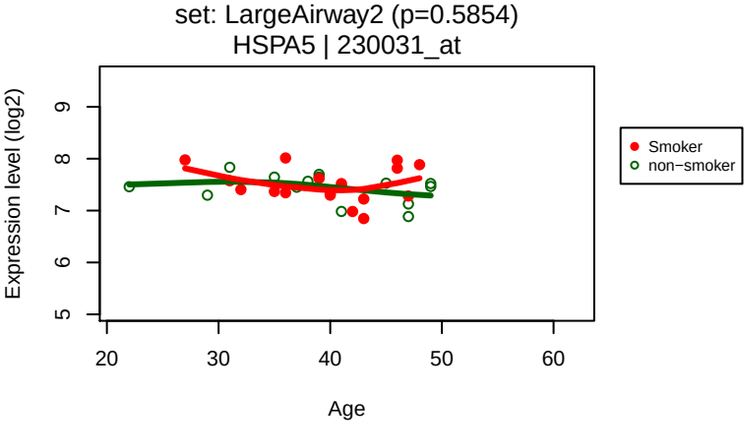

Gene expression ratio (smokers/nonsmokers)
gene: HSPA5 | 230031_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5707
p=0.6177
p=0.7587
p=0.0667
p=0.5854
p=0.5508
p=0.2484
p=0.6914
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

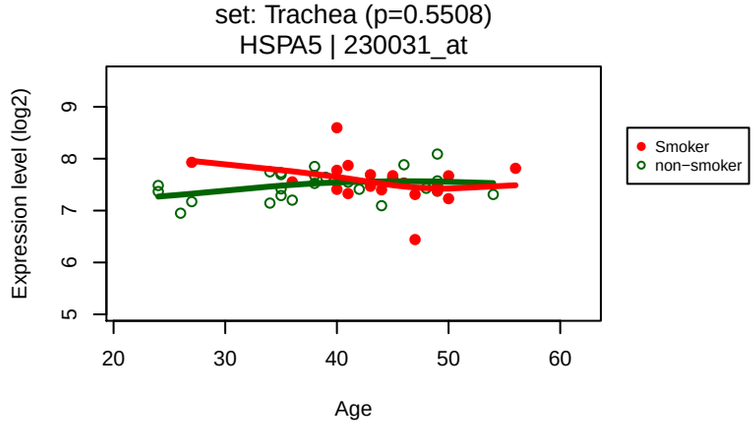

Gene expression ratio in Males
gene: HSPA5 | 230031_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7558
p=0.9656
p=0.7945
p=0.4918
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA5 | 230031_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8109
p=0.5083
p=0.9736
p=0.8188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

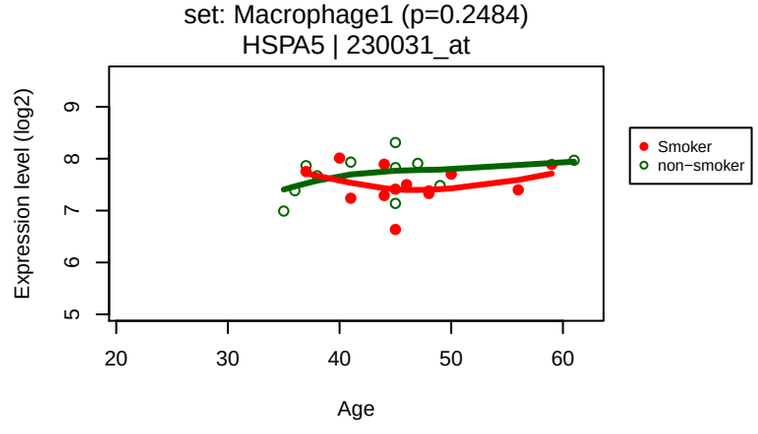

set: SmallAirway1 (p=0.5707)
HSPA5 | 230031_at
Expression level (log2)
Age
Smoker
non−smoker

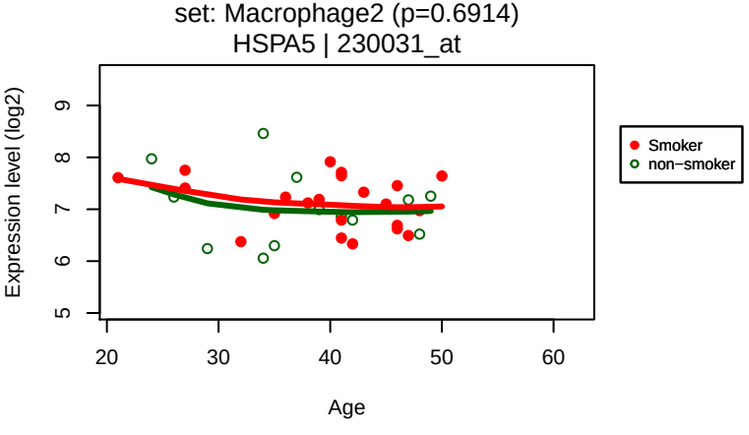

set: SmallAirway2 (p=0.6177)
HSPA5 | 230031_at
Expression level (log2)
Age
Smoker
non−smoker

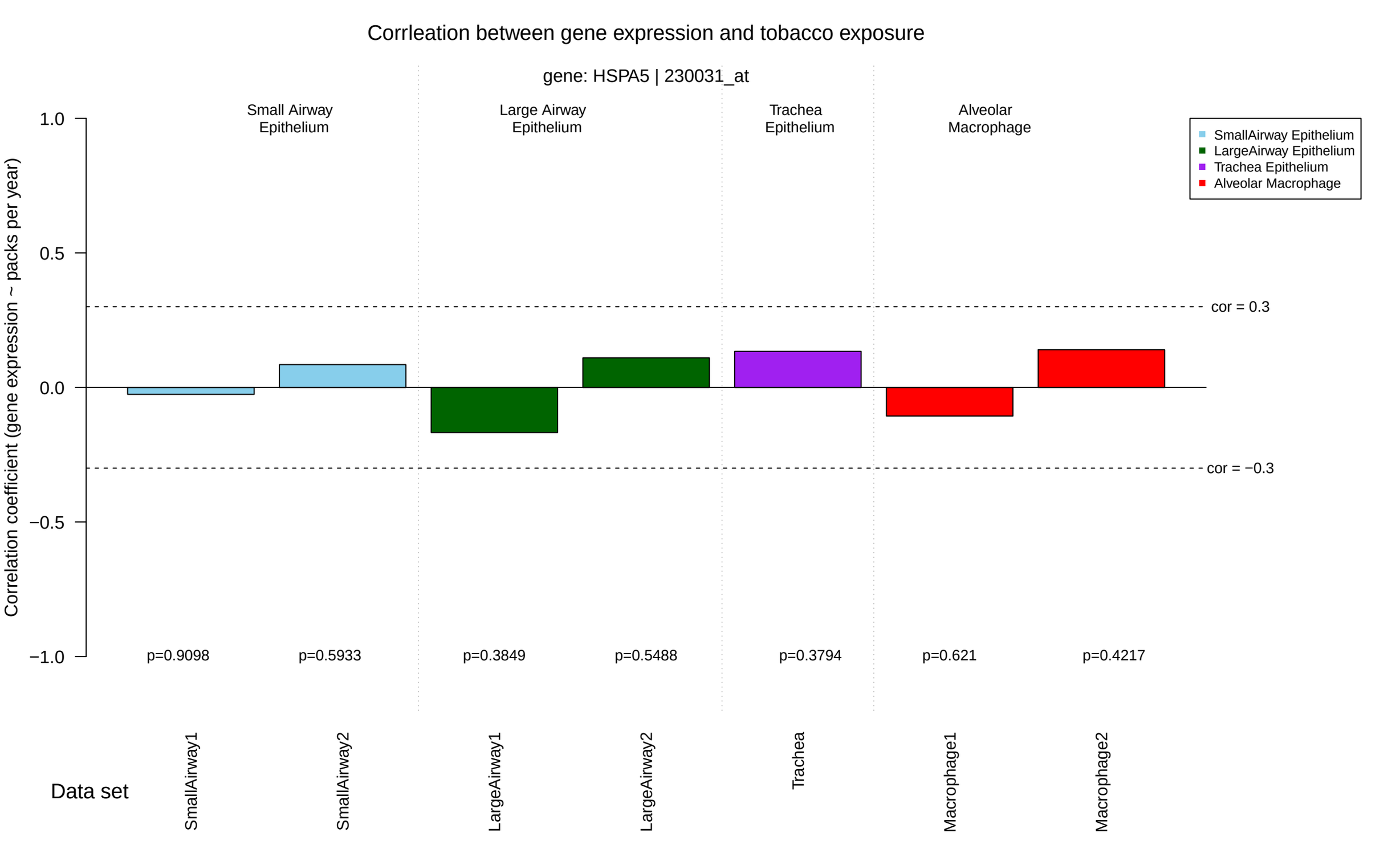

set: LargeAirway1 (p=0.0667)
HSPA5 | 230031_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5854)
HSPA5 | 230031_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5508)
HSPA5 | 230031_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2484)
HSPA5 | 230031_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6914)
HSPA5 | 230031_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA5 | 230031_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9098
p=0.5933
p=0.3849
p=0.5488
p=0.3794
p=0.621
p=0.4217
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2